Local Sports League Management System

Final database demonstration

Group 13

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CS-3743 Database Systems

Presentation Roadmap

1. **Business summary** — Why local sport leagues needs the database
2. **Rules + ERD/DDD** — How the data is organized and normalized
3. **Sample data** — How repeated real-world records are handled
4. **Query demo** — Fourteen reports/actions that answer business questions
5. **Reflection** — What we learned and how the system delivers value

Business Summary: What the League Needs

Client

Local community sports league organization
Manages recreational sports across seasons
Tracks teams, players, coaches, matches, locations, and statistics

Current problem

Information is spread across manual records and spreadsheets
Schedules, rosters, results, and standings can become inconsistent
Staff need faster access to accurate reports

Database benefit

One reliable place to manage schedules, rosters, standings, and player statistics.

Reduces duplicate work and data entry errors
Improves communication with coaches, players, and families
Supports planning for facilities, staffing, and future leagues

Business Questions the Database Answers

Participation

How many teams and players are in each sport or season?

Rosters

Which players and coaches are assigned to each team?

Scheduling

What matches are scheduled, and where will they be played?

Results

Which matches are completed, and what were the final scores?

Standings

How do wins, losses, ties, and points compare by team?

Player stats

Which players lead in goals, assists, fouls, or minutes?

These questions guide the ERD, sample data, and query demonstrations.

Business Rules: League Structure

Sport → Season

A sport can have many seasons; each season belongs to one sport.

Season ↔ Team

A season can include many teams, and teams can return across seasons.

Team-season record

A team participates at most once per season and includes division/age group.

Player assignment

A player may participate in many seasons but only one team per season.

Roster record

Each player-season assignment references exactly one team-season record.

Coach assignment

A coach may work across seasons; each team-season has one assigned coach.

Bridge tables let the same teams, players, and coaches appear across seasons without duplicating core records.

Business Rules: Matches, Results, and Controls

Match scheduling
Each match belongs to one season and has a valid date, time, and location.

Standings
Wins, losses, ties, and points are tracked by team-season.

Home + away teams
Each match has exactly two different teams from the season.

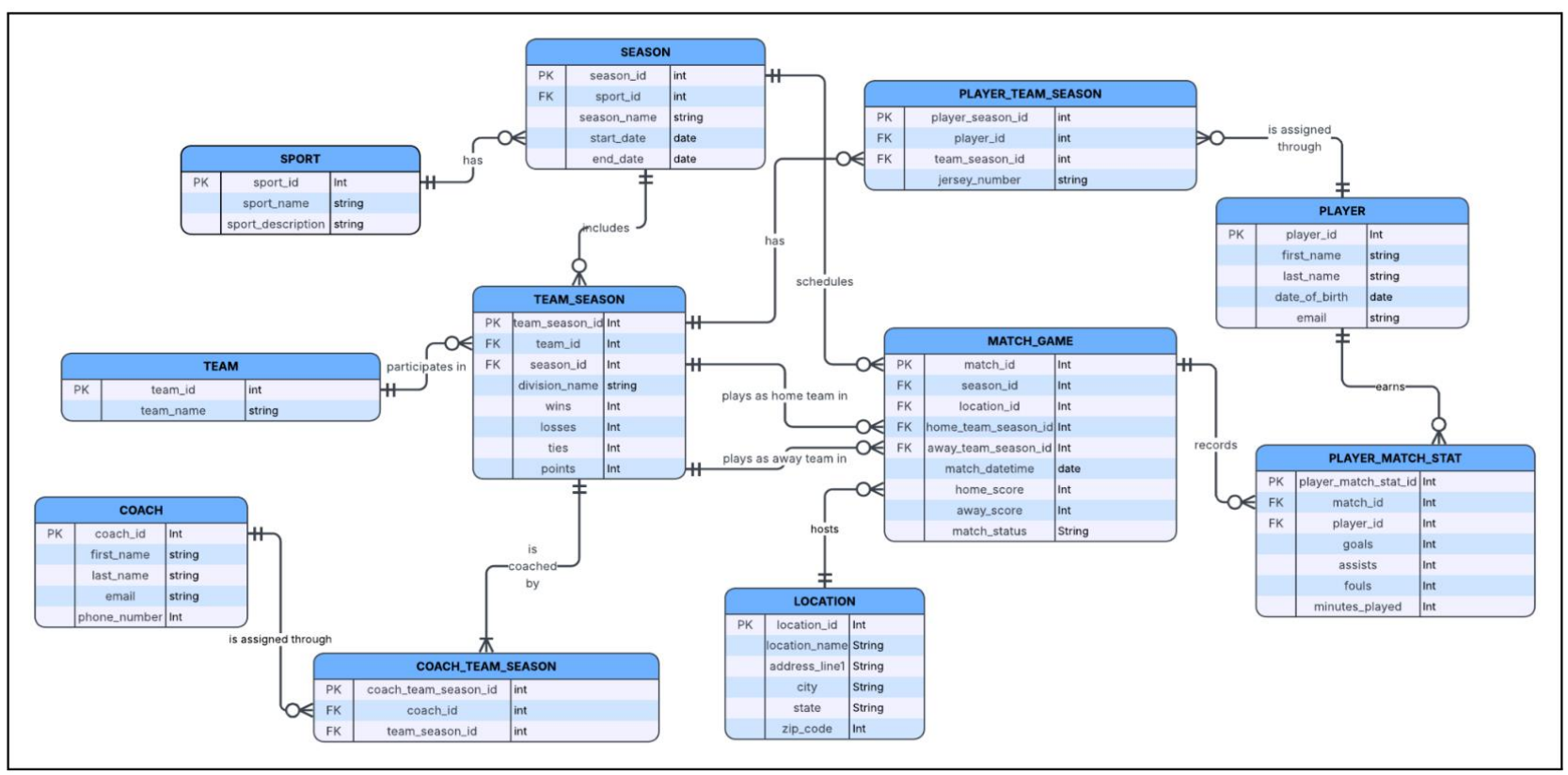
Player stats
A player can have at most one stat record per match.

Status and scores
Scheduled matches may exist before scores are recorded.

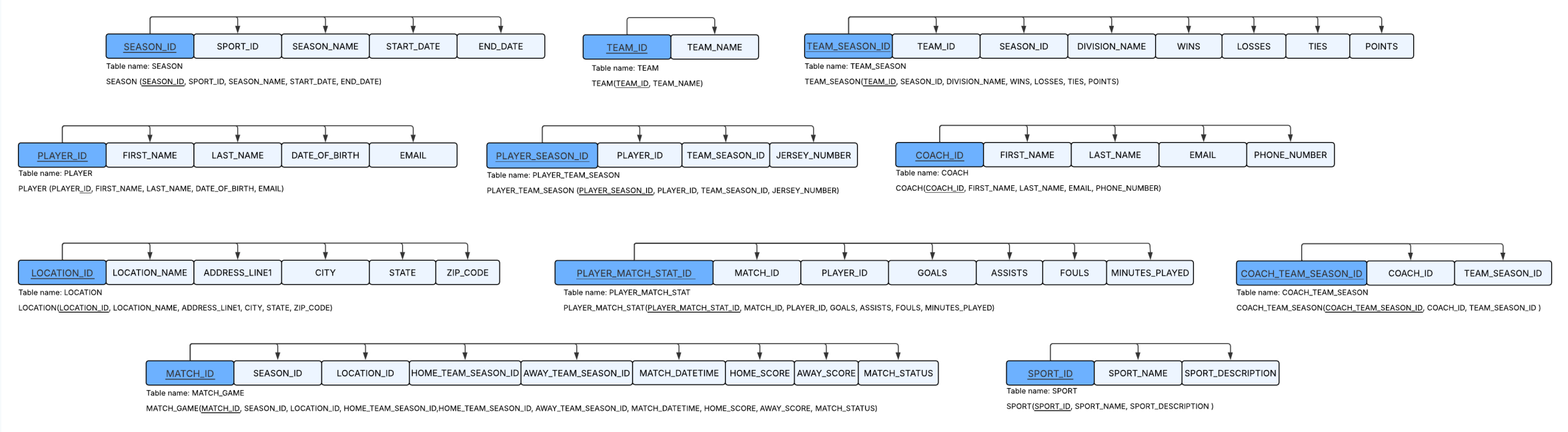
No double-booking
A location cannot host more than one match at the same date/time.

These rules become database safeguards: keys, foreign keys, unique constraints, and checks.

Final ERD


SEASON
PK season_id int
FK sport_id int
season_name string
start_date date
end_date date
SPORT
PK sport_id Int
sport_name string
sport_description string
has
includes
PLAYER_TEAM_SEASON
PK player_season_id int
FK player_id int
FK team_season_id int
jersey_number string
is assigned through
PLAYER
PK player_id Int
first_name string
last_name string
date_of_birth date
email string
has
schedules
TEAM_SEASON
PK team_season_id Int
FK team_id Int
FK season_id Int
division_name string
wins Int
losses Int
ties Int
points Int
TEAM
PK team_id int
team_name string
participates in
MATCH_GAME
PK match_id Int
FK season_id Int
FK location_id Int
FK home_team_season_id Int
FK away_team_season_id Int
match_datetime date
home_score Int
away_score Int
match_status String
plays as home team in
plays as away team in
earns
records
PLAYER_MATCH_STAT
PK player_match_stat_id Int
FK match_id Int
FK player_id Int
goals Int
assists Int
fouls Int
minutes_played Int
COACH
PK coach_id Int
first_name string
last_name string
email string
phone_number Int
is coached by
hosts
LOCATION
PK location_id Int
location_name String
address_line1 String
city String
state String
zip_code Int
is assigned through
COACH_TEAM_SEASON
PK coach_team_season_id int
FK coach_id int
FK team_season_id int

Final DDD and Normalization



3rd Normal Form achieved: Yes — tables separate core entities from season assignments, rosters, coach assignments, matches, locations, and player stats.

Sample Data Coverage

10 Sports	**16** Seasons	**24** Teams	**48** Players	**24** Coaches
15 Locations	**24** Team-season records	**48** Roster assignments	**30** Matches	**96** Player stat rows

The data set is large enough to test schedules, rosters, standings, statistics, and reporting across multiple sports.

Handling Repetition Without Duplicating Records

Example 1: The same team appears in multiple matches

match_id	match_datetime	home_team	away_team	home_score	away_score	match_status
901	2025-03-08 10:00:00	Alamo Strikers	Cactus Kickers	3	1	Completed
903	2025-03-22 10:00:00	Alamo Strikers	Hilltop Hurricanes	1	1	Completed
905	2025-04-05 10:00:00	Alamo Strikers	River Runners	NULL	NULL	Scheduled

Example 2: The same team has multiple top scorers

team_name	first_name	last_name	total_goals
River Runners	Elijah	Perez	1
River Runners	Ava	Castillo	1

Normalized design stores the team and player once, then connects repeated activity through MATCH_GAME and PLAYER_MATCH_STAT.

Query 1: Season Scoring Summary

COUNT + SUM + AVG

Business rationale

Shows how active each season is
Compares total scoring across sports
Supports resource planning by sport or season

Real-world use

League managers can identify high-activity seasons and plan staffing, scorekeeping, and facility needs.

Season	Sport	Total_Matches	Total_Points_Scored	Avg_Points_Per_Match
Winter Basketball 2025	Basketball	5	429	107.25
Summer Flag Football 2025	Flag Football	5	169	42.25
Spring Baseball 2025	Baseball	5	31	7.75
Fall Volleyball 2025	Volleyball	5	17	4.25
Spring Soccer 2025	Soccer	5	12	3.00
Spring Tennis 2025	Tennis	5	11	2.75
Fall Baseball 2025	Baseball	0	0	0.00
Fall Soccer 2025	Soccer	0	0	0.00
Fall Volleyball 2024	Volleyball	0	0	0.00
Spring Rugby 2026	Rugby	0	0	0.00
Spring Tennis 2026	Tennis	0	0	0.00
Summer Basketball 2025	Basketball	0	0	0.00
Summer Flag Football 2026	Flag Football	0	0	0.00
Swimming Season 2025	Swimming	0	0	0.00
Track Season 2025	Track	0	0	0.00
Winter Hockey 2025	Hockey	0	0	0.00

Key output: Winter Basketball has the highest total scoring activity.

Query 2: Unique Coach Assignments

JOIN + CONCAT + DISTINCT

Business rationale

Lists unique coaches, teams, and seasons
Converts IDs into readable names
Prevents confusion over assignments

Real-world use

Administrative staff can confirm coach coverage before seasons begin and quickly find who is responsible for each team.

coach_name	team_name	season_name
Andre Coleman	First Down Force	Summer Flag Football 2025
Angela Price	River Runners	Spring Soccer 2025
Bianca Wells	Court Commanders	Spring Tennis 2025
Brandon Lopez	Hoop Hustlers	Winter Basketball 2025
Chris Evans	Volley Kings	Spring Tennis 2025
Daniel Reed	Alamo Strikers	Spring Soccer 2025
Evan Brooks	Pitch Perfect	Spring Baseball 2025
Hector Mendez	Blitz Brigade	Summer Flag Football 2025
Jasmine Young	Goal Line Guardians	Summer Flag Football 2025
Jorge Villarreal	Spike Squad	Fall Volleyball 2025
Karen Flores	Cactus Kickers	Spring Soccer 2025
Kelly Watson	Net Breakers	Fall Volleyball 2025
Luis Garza	Diamond Dynamos	Spring Baseball 2025
Marcus Hill	Hilltop Hurricanes	Spring Soccer 2025
Maribel Rojas	Smash Squad	Spring Tennis 2025
Monica Salinas	Home Run Heroes	Spring Baseball 2025
Nina Patel	Block Party	Fall Volleyball 2025
Omar Khan	Serve Aces	Fall Volleyball 2025
Patrick Owen	Racket Rebels	Spring Tennis 2025
Priya Shah	Red Zone Raiders	Summer Flag Football 2025
Rachel Nguyen	Fast Breakers	Winter Basketball 2025
Sofia Diaz	Backboard Bandits	Winter Basketball 2025
Tanya Morris	Base Blazers	Spring Baseball 2025
Victor Santos	Court Kings	Winter Basketball 2025

Key output: each coach is tied to a readable team and season.

Query 3: Teams With No Losses

Subquery + UNION

Business rationale

Identifies undefeated teams
Supports standings and awards
Checks both home and away losses

Real-world use

League leadership can quickly find standout teams, support playoff decisions, and publish standings updates.

team_name
Alamo Strikers
Base Blazers
Blitz Brigade
Court Commanders
Court Kings
Diamond Dynamos
Goal Line Guardians
Hilltop Hurricanes
Net Breakers
Racket Rebels
Spike Squad

Key output: teams returned never appear on the losing side of a completed match.

Query 4: Match Results by Most Recent Date

JOIN + ORDER BY

Business rationale

Shows recent matches first
Uses team names instead of IDs
Combines schedules and results

Real-world use

Families, coaches, and staff can see the latest game outcomes and upcoming scheduled games.

Key output: scheduled games can appear before scores are entered.

match_id	match_datetime	home_team	away_team	home_score	away_score	match_status
920	2025-10-04 12:00:00	Spike Squad	Net Breakers	NULL	NULL	Scheduled
919	2025-09-27 14:00:00	Block Party	Net Breakers	1	3	Completed
918	2025-09-20 12:00:00	Spike Squad	Serve Aces	3	2	Completed
917	2025-09-13 14:00:00	Serve Aces	Net Breakers	2	3	Completed
916	2025-09-06 12:00:00	Spike Squad	Block Party	3	0	Completed
925	2025-07-05 18:00:00	Blitz Brigade	Goal Line Guardians	NULL	NULL	Scheduled
924	2025-06-28 20:00:00	Red Zone Raiders	Goal Line Guardians	12	18	Completed
923	2025-06-21 18:00:00	Blitz Brigade	First Down Force	35	20	Completed
922	2025-06-14 20:00:00	First Down Force	Goal Line Guardians	21	21	Completed
921	2025-06-07 18:00:00	Blitz Brigade	Red Zone Raiders	28	14	Completed
915	2025-05-03 16:00:00	Diamond Dynamos	Base Blazers	NULL	NULL	Scheduled
914	2025-04-26 18:00:00	Home Run Heroes	Base Blazers	1	7	Completed
913	2025-04-19 16:00:00	Diamond Dynamos	Pitch Perfect	5	2	Completed
912	2025-04-12 18:00:00	Pitch Perfect	Base Blazers	3	3	Completed
930	2025-04-12 09:00:00	Racket Rebels	Court Commanders	NULL	NULL	Scheduled
911	2025-04-05 16:00:00	Diamond Dynamos	Home Run Heroes	6	4	Completed
929	2025-04-05 11:00:00	Smash Squad	Court Commanders	0	2	Completed
905	2025-04-05 10:00:00	Alamo Strikers	River Runners	NULL	NULL	Scheduled
904	2025-03-29 12:00:00	Cactus Kickers	River Runners	2	0	Completed
928	2025-03-29 09:00:00	Racket Rebels	Volley Kings	2	2	Completed
927	2025-03-22 11:00:00	Volley Kings	Court Commanders	1	2	Completed
903	2025-03-22 10:00:00	Alamo Strikers	Hilltop Hurricanes	1	1	Completed
902	2025-03-15 12:00:00	Hilltop Hurricanes	River Runners	2	2	Completed
926	2025-03-15 09:00:00	Racket Rebels	Smash Squad	2	0	Completed
901	2025-03-08 10:00:00	Alamo Strikers	Cactus Kickers	3	1	Completed
910	2025-02-08 14:00:00	Court Kings	Backboard Bandits	NULL	NULL	Scheduled
909	2025-02-01 16:00:00	Fast Breakers	Backboard Bandits	55	48	Completed
908	2025-01-25 14:00:00	Court Kings	Hoop Hustlers	61	61	Completed
907	2025-01-18 16:00:00	Hoop Hustlers	Backboard Bandits	45	49	Completed
906	2025-01-11 14:00:00	Court Kings	Fast Breakers	58	52	Completed

Query 5: Readable Match Results Report

JOIN + CONCAT score

Business rationale

Shows date, location, teams, and score
Creates a user-friendly final score field
Separates scheduled and completed matches

Real-world use

League staff can publish or review match reports without looking across several tables.

Key output: match details are combined into one readable report.

Match_ID	Match_Date	Location	Home_Team	Away_Team	Final_Score	Match_Status
906	2025-01-11	Southside Fields	Court Kings	Fast Breakers	58 - 52	Completed
907	2025-01-18	Westside Courts	Hoop Hustlers	Backboard Bandits	45 - 49	Completed
908	2025-01-25	Alamo City Arena	Court Kings	Hoop Hustlers	61 - 61	Completed
909	2025-02-01	Hill Country Sports Park	Fast Breakers	Backboard Bandits	55 - 48	Completed
910	2025-02-08	Downtown Tennis Center	Court Kings	Backboard Bandits	NULL	Scheduled
901	2025-03-08	Northside Sports Complex	Alamo Strikers	Cactus Kickers	3 - 1	Completed
926	2025-03-15	Leon Valley Athletic Center	Racket Rebels	Smash Squad	2 - 0	Completed
902	2025-03-15	Community Field West	Hilltop Hurricanes	River Runners	2 - 2	Completed
903	2025-03-22	River Park Gym	Alamo Strikers	Hilltop Hurricanes	1 - 1	Completed
927	2025-03-22	Stone Oak Sports Pavilion	Volley Kings	Court Commanders	1 - 2	Completed
928	2025-03-29	Medical Center Courts	Racket Rebels	Volley Kings	2 - 2	Completed
904	2025-03-29	Eastside Recreation Center	Cactus Kickers	River Runners	2 - 0	Completed
905	2025-04-05	Mission Baseball Park	Alamo Strikers	River Runners	NULL	Scheduled
929	2025-04-05	Brooks City Base Fields	Smash Squad	Court Commanders	0 - 2	Completed
911	2025-04-05	Leon Valley Athletic Center	Diamond Dynamos	Home Run Heroes	6 - 4	Completed
930	2025-04-12	Helotes Community Park	Racket Rebels	Court Commanders	NULL	Scheduled
912	2025-04-12	Stone Oak Sports Pavilion	Pitch Perfect	Base Blazers	3 - 3	Completed
913	2025-04-19	Medical Center Courts	Diamond Dynamos	Pitch Perfect	5 - 2	Completed
914	2025-04-26	Brooks City Base Fields	Home Run Heroes	Base Blazers	1 - 7	Completed
915	2025-05-03	Helotes Community Park	Diamond Dynamos	Base Blazers	NULL	Scheduled
921	2025-06-07	Southside Fields	Blitz Brigade	Red Zone Raiders	28 - 14	Completed
922	2025-06-14	Westside Courts	First Down Force	Goal Line Guardians	21 - 21	Completed
923	2025-06-21	Alamo City Arena	Blitz Brigade	First Down Force	35 - 20	Completed
924	2025-06-28	Hill Country Sports Park	Red Zone Raiders	Goal Line Guardians	12 - 18	Completed
925	2025-07-05	Downtown Tennis Center	Blitz Brigade	Goal Line Guardians	NULL	Scheduled
916	2025-09-06	Northside Sports Complex	Spike Squad	Block Party	3 - 0	Completed
917	2025-09-13	Community Field West	Serve Aces	Net Breakers	2 - 3	Completed
918	2025-09-20	River Park Gym	Spike Squad	Serve Aces	3 - 2	Completed
919	2025-09-27	Eastside Recreation Center	Block Party	Net Breakers	1 - 3	Completed
920	2025-10-04	Mission Baseball Park	Spike Squad	Net Breakers	NULL	Scheduled

Query 6: Coach Assignment With Contact Info

JOIN across assignment tables

Business rationale

Shows coach, team, season, and division
Includes phone number for contact needs
Supports staffing and communication

Real-world use

Administrators can contact the correct coach for schedule changes, roster issues, or match updates.

Coach_Name	Team	Season	Division	Coach_Phone
Nina Patel	Block Party	Fall Volleyball 2025	Open Volleyball	210-555-1014
Kelly Watson	Net Breakers	Fall Volleyball 2025	Open Volleyball	210-555-1016
Omar Khan	Serve Aces	Fall Volleyball 2025	Open Volleyball	210-555-1015
Jorge Villarreal	Spike Squad	Fall Volleyball 2025	Open Volleyball	210-555-1013
Tanya Morris	Base Blazers	Spring Baseball 2025	City Baseball	210-555-1012
Luis Garza	Diamond Dynamos	Spring Baseball 2025	City Baseball	210-789-1009
Monica Salinas	Home Run Heroes	Spring Baseball 2025	City Baseball	210-555-1010
Evan Brooks	Pitch Perfect	Spring Baseball 2025	City Baseball	210-555-1011
Daniel Reed	Alamo Strikers	Spring Soccer 2025	U14 Soccer	210-533-1001
Karen Flores	Cactus Kickers	Spring Soccer 2025	U14 Soccer	210-235-1002
Marcus Hill	Hilltop Hurricanes	Spring Soccer 2025	U14 Soccer	210-895-1003
Angela Price	River Runners	Spring Soccer 2025	U14 Soccer	210-547-1004
Bianca Wells	Court Commanders	Spring Tennis 2025	Tennis Singles	210-555-1024
Patrick Owen	Racket Rebels	Spring Tennis 2025	Tennis Singles	210-555-1021
Maribel Rojas	Smash Squad	Spring Tennis 2025	Tennis Singles	210-555-1022
Chris Evans	Volley Kings	Spring Tennis 2025	Tennis Singles	210-555-1023
Hector Mendez	Blitz Brigade	Summer Flag Footb...	Flag Football	210-555-1017
Andre Coleman	First Down Force	Summer Flag Footb...	Flag Football	210-555-1019
Jasmine Young	Goal Line Guardians	Summer Flag Footb...	Flag Football	210-555-1020
Priya Shah	Red Zone Raiders	Summer Flag Footb...	Flag Football	210-555-1018
Sofia Diaz	Backboard Bandits	Winter Basketball 2...	Junior Basketball	210-467-1008
Victor Santos	Court Kings	Winter Basketball 2...	Junior Basketball	210-231-1005
Rachel Nguyen	Fast Breakers	Winter Basketball 2...	Junior Basketball	210-093-1006
Brandon Lopez	Hoop Hustlers	Winter Basketball 2...	Junior Basketball	210-890-1007

Key output: each team-season has a clear coach contact.

Query 7: Total Goals by Player

SUM + GROUP BY

Business rationale

Ranks players by total goals
Summarizes multiple stat records
Supports awards and performance tracking

Real-world use

Coaches and league staff can identify top scorers for recognition, standings articles, or player development discussions.

Key output: Lucas Morales leads the shown sample with 60 goals.

player_id	player_name	total_goals
309	Lucas Morales	60
310	Mia Flores	59
313	Logan Gonzalez	54
311	Ethan Sanchez	54
312	Isabella Ramirez	53
314	Camila Reyes	52
315	James Gutierrez	49
316	Amelia Vega	48
333	Isaac Valdez	32
334	Nora Fuentes	31
337	Gabriel Nunez	21
339	Adrian Molina	20
338	Aria Herrera	20
340	Zoe Cardenas	19
336	Layla Acosta	13
335	Julian Cabrera	13
323	Samuel Navarro	6
317	Benjamin Cruz	6
318	Harper Ortiz	5
331	Leo Pena	4
325	David Vargas	4
324	Sofia Aguilar	4
319	Daniel Mendoza	3
321	Mateo Soto	3
301	Liam Garcia	3
329	Anthony Delgado	2
330	Grace Rios	2
326	Victoria Medina	2
332	Chloe Salazar	2
322	Abigail Jimenez	2

Query 8: Matches Sorted by Date

ORDER BY date descending

Business rationale

Provides a quick schedule view
Shows match status beside each date
Places newest or upcoming items first

Real-world use

Staff can quickly check the most recent schedule entries and confirm whether matches are scheduled or completed.

match_id	match_datetime	match_status
920	2025-10-04 12:00:00	Scheduled
919	2025-09-27 14:00:00	Completed
918	2025-09-20 12:00:00	Completed
917	2025-09-13 14:00:00	Completed
916	2025-09-06 12:00:00	Completed
925	2025-07-05 18:00:00	Scheduled
924	2025-06-28 20:00:00	Completed
923	2025-06-21 18:00:00	Completed
922	2025-06-14 20:00:00	Completed
921	2025-06-07 18:00:00	Completed
915	2025-05-03 16:00:00	Scheduled
914	2025-04-26 18:00:00	Completed
913	2025-04-19 16:00:00	Completed
912	2025-04-12 18:00:00	Completed
930	2025-04-12 09:00:00	Scheduled
911	2025-04-05 16:00:00	Completed
929	2025-04-05 11:00:00	Completed
905	2025-04-05 10:00:00	Scheduled
904	2025-03-29 12:00:00	Completed
928	2025-03-29 09:00:00	Completed
927	2025-03-22 11:00:00	Completed
903	2025-03-22 10:00:00	Completed
902	2025-03-15 12:00:00	Completed
926	2025-03-15 09:00:00	Completed
901	2025-03-08 10:00:00	Completed
910	2025-02-08 14:00:00	Scheduled
909	2025-02-01 16:00:00	Completed
908	2025-01-25 14:00:00	Completed
907	2025-01-18 16:00:00	Completed
906	2025-01-11 14:00:00	Completed

Key output: latest scheduled volleyball match appears at the top.

Query 9: Find the Top Goal Scorer for each Team

Multi-table
JOIN + Aggregation + Correlated Subquery

Business rationale

Aggregates goals by player
Supports player evaluation
Orders by team name

Real-world use

Coaches can evaluate offensive production and identify key offensive players for each team.

team_name	first_name	last_name	total_goals
Smash Squad	Sebastian	Pacheco	0
Smash Squad	Aurora	Montoya	0
River Runners	Elijah	Perez	1
River Runners	Ava	Castillo	1
Block Party	Joseph	Silva	1
Cactus Kickers	Noah	Martinez	2
Hilltop Hurricanes	Mason	Rivera	2
Serve Aces	Anthony	Delgado	2
Serve Aces	Grace	Rios	2
Racket Rebels	Aaron	Espinoza	2
Racket Rebels	Natalie	Rosales	2
Volley Kings	Thomas	Ibarra	2
Court Commanders	Caleb	Quintero	2
Court Commanders	Maya	Benitez	2
Alamo Strikers	Liam	Garcia	3
Home Run Heroes	Daniel	Mendoza	3
Pitch Perfect	Mateo	Soto	3
Spike Squad	David	Vargas	4
Net Breakers	Leo	Pena	4
Diamond Dynamos	Benjamin	Cruz	6
Base Blazers	Samuel	Navarro	6
Red Zone Raiders	Julian	Cabrera	13
Red Zone Raiders	Layla	Acosta	13
Goal Line Guardi...	Adrian	Molina	20
First Down Force	Gabriel	Nunez	21
Blitz Brigade	Isaac	Valdez	32
Backboard Bandits	James	Gutierrez	49
Fast Breakers	Ethan	Sanchez	54
Hoop Hustlers	Logan	Gonzalez	54
Court Kings	Lucas	Morales	60

Key output: Lists top scorer for each team, orders results from lowest to highest total goals

Query 10: Player Team Roster

Multi-Table Join

Business rationale

Lists players by assigned team
Includes jersey numbers
Orders by team name

Real-world use

Coaches and administrators can verify rosters before games and correct player assignment issues.

Key output: each roster row connects a player to a team and jersey number.

team_name	first_name	last_name	jersey_number
Alamo Strikers	Liam	Garcia	2
Alamo Strikers	Emma	Lopez	4
Backboard Bandits	James	Gutierrez	20
Backboard Bandits	Amelia	Vega	21
Base Blazers	Sofia	Aguilar	33
Base Blazers	Samuel	Navarro	31
Blitz Brigade	Nora	Fuentes	3
Blitz Brigade	Isaac	Valdez	1
Block Party	Luna	Campos	42
Block Party	Joseph	Silva	40
Cactus Kickers	Sophia	Hernandez	7
Cactus Kickers	Noah	Martinez	5
Court Commanders	Maya	Benitez	46
Court Commanders	Caleb	Quintero	43
Court Kings	Mia	Flores	13
Court Kings	Lucas	Morales	12
Diamond Dynamos	Benjamin	Cruz	22
Diamond Dynamos	Harper	Ortiz	23
Fast Breakers	Isabella	Ramirez	16
Fast Breakers	Ethan	Sanchez	15
First Down Force	Aria	Herrera	25
First Down Force	Gabriel	Nunez	17
Goal Line Guardians	Zoe	Cardenas	29
Goal Line Guardians	Adrian	Molina	26
Hilltop Hurricanes	Mason	Rivera	8
Hilltop Hurricanes	Olivia	Torres	9
Home Run Heroes	Daniel	Mendoza	24
Home Run Heroes	Evelyn	Ramos	27
Hoop Hustlers	Logan	Gonzalez	18
Hoop Hustlers	Camila	Reyes	19
Net Breakers	Leo	Pena	50

Query 11: Total Minutes Played by Player

SUM + GROUP BY

Business rationale

Tracks player participation
Sums minutes across matches
Helps monitor playing time balance

Real-world use

Coaches can review whether players are receiving reasonable playing time across the season.

player_id	player_name	total_minutes_played
321	Mateo Soto	140
322	Abigail Jimenez	140
323	Samuel Navarro	140
324	Sofia Aguilar	140
333	Isaac Valdez	120
334	Nora Fuentes	120
335	Julian Cabrera	120
336	Layla Acosta	120
337	Gabriel Nunez	120
338	Aria Herrera	120
339	Adrian Molina	120
340	Zoe Cardenas	120
325	David Vargas	120
326	Victoria Medina	120
327	Joseph Silva	120
328	Luna Campos	120
329	Anthony Delgado	120
330	Grace Rios	120
331	Leo Pena	120
332	Chloe Salazar	120
309	Lucas Morales	96

Key output: players with the most recorded minutes appear first.

Query 12: Average Score by Location

AVG + GROUP BY location

Business rationale

Compares scoring by venue
Highlights high-scoring locations
Supports scheduling analysis

Real-world use

League staff can evaluate whether certain locations tend to host higher-scoring matches or need special planning.

location_name	avg_total_score
Alamo City Arena	88.50
Southside Fields	76.00
Westside Courts	68.00
Hill Country Sports Park	66.50
Leon Valley Athletic Center	6.00
Medical Center Courts	5.50
Brooks City Base Fields	5.00
Community Field West	4.50
Stone Oak Sports Pavilion	4.50
Northside Sports Complex	3.50
River Park Gym	3.50
Eastside Recreation Center	3.00
Mission Baseball Park	NULL
Downtown Tennis Center	NULL
Helotes Community Park	NULL

Key output: Alamo City Arena has the highest average total score shown.

Query 13: Insert Trigger Updates Team Stats

AFTER INSERT trigger

Business rationale

Automatically updates wins, losses, ties, and points
Reduces manual standings errors
Keeps standings synchronized with completed matches

Example action

Insert a completed match where Alamo Strikers defeat Cactus Kickers, 2–1.

Before trigger

team_season	team_name	wins	losses	ties	points
601	Alamo Strikers	1	0	1	4
602	Cactus Kickers	1	1	0	3

After trigger

team_season	team_name	wins	losses	ties	points
601	Alamo Strikers	2	0	1	7
602	Cactus Kickers	1	2	0	3

Key output: standings change automatically when a new completed match is inserted.

Query 14: Delete Trigger Reverses Team Stats

AFTER DELETE trigger

Business rationale

Reverses standings when a match is removed
Protects reports from stale totals
Supports correction of mistaken match entries

Example action

Delete the inserted match; the trigger subtracts the related win, loss, and points.

Before delete

team_season	team_name	wins	losses	ties	points
601	Alamo Strikers	2	0	1	7
602	Cactus Kickers	1	2	0	3

After delete trigger

team_season	team_name	wins	losses	ties	points
601	Alamo Strikers	1	0	1	4
602	Cactus Kickers	1	1	0	3

Key output: standings return to their previous values after the match is deleted.

Team Reflection: Objectives, Challenges, and Solutions

Overall: The team met the original objectives by building a centralized sports league database that supports sports, seasons, teams, players, coaches, schedules, locations, match results, standings, and player statistics while reducing repetition through bridge tables and constraints.

Cameron Bergeron

Season activity & match results

My main reflection is that the database successfully met the goal of helping league organizers view season activity and match results in a clear way. One of the biggest challenges was making sure the database could track teams across different seasons without repeating the same team information over and over. We solved this by using the `TEAM_SEASON` table, which connects each team to a specific season and stores season-based information like division, wins, losses, ties, and points.

Bryan Vera

Player and team performance

My reflection is that the database met the objective of answering performance-related questions, especially through player and team statistics. The most significant challenge was making sure player stats were tied to the correct match without creating duplicate stat records. We solved this by creating the `PLAYER_MATCH_STAT` table and using a unique relationship between each player and match, so each player can only have one stat record per match.

Dalton J. Opudo

Coach, team, and season assignments

My reflection is that the database supports the business rule that coaches, teams, and seasons must be connected in an organized way. A challenge was making sure coach assignments were clear because coaches may work with different teams across different seasons. We solved this by using the `COACH_TEAM_SEASON` table, which connects a coach to a team's specific season record instead of only connecting the coach directly to a team.

Lance Lamon Martin

Scheduling and location control

My reflection is that the database met the scheduling objective by allowing league staff to see when and where matches are played. The biggest challenge was preventing scheduling conflicts, especially if two matches were accidentally assigned to the same location at the same time. We solved this by including a rule in the database that makes the combination of `location_id` and `match_datetime` unique, which helps prevent double-booking.

Closing: Value Delivered to Users

The database turns league activity into organized, reliable, and useful information.

Centralized records

Teams, players, coaches, seasons, locations, matches, and stats are stored consistently.

User-focused reports

Queries answer real questions about participation, schedules, results, standings, and performance.

Better decisions

League staff can plan facilities, staffing, standings, and future seasons with confidence.